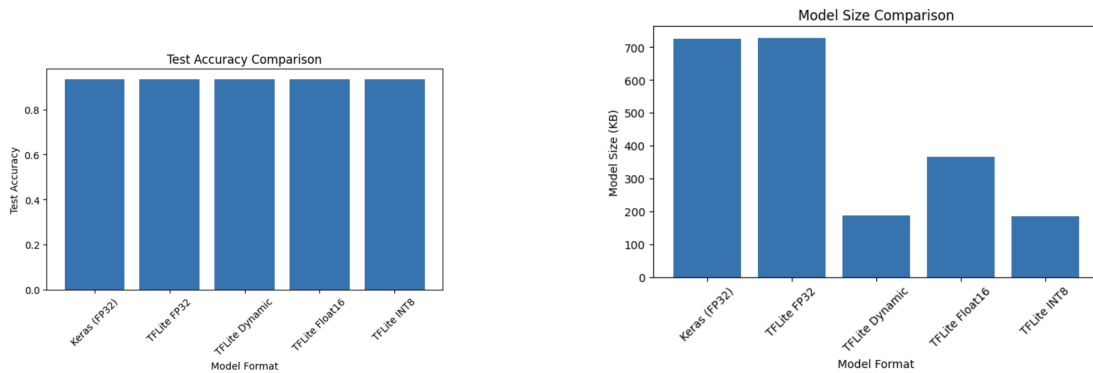
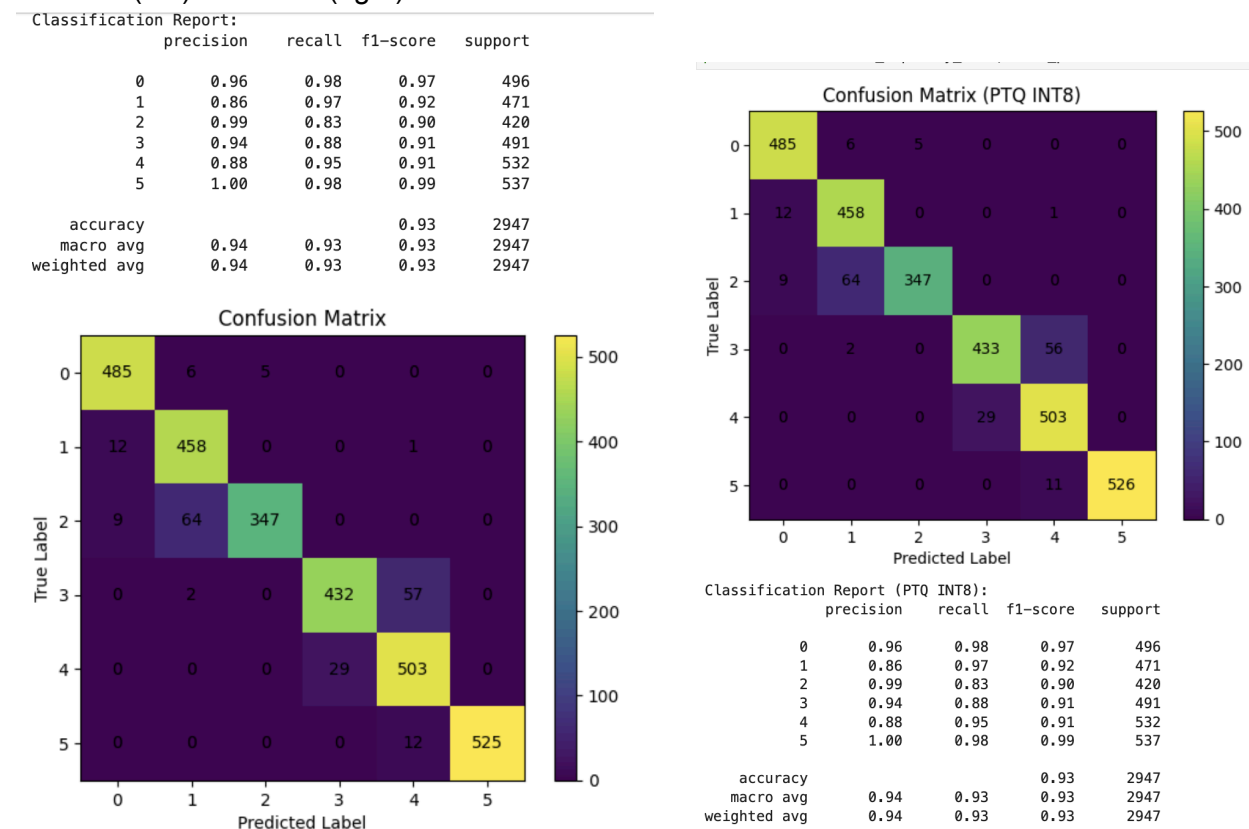


Quantization of a DNN Using the UCI Human Activity Recognition Dataset



Confusion matrix

Baseline (left) and INT8 (right)

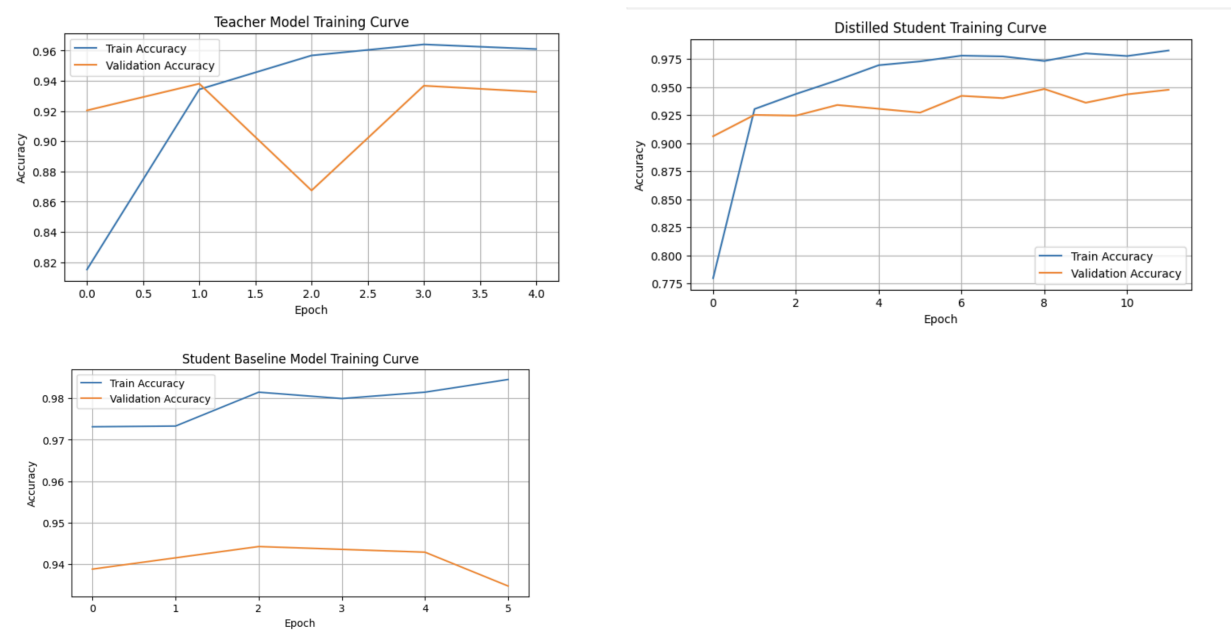


Short summary question

1. The INT8 quantization produced the smallest model size because weights and activations are stored using 8-bit integers.
2. The FP32 TensorFlow Lite model achieved the best accuracy since it preserves full precision.

- Yes, QAT improves the accuracy compared to PTQ INT8 because the model was trained with quantization effects included, allowing it to adapt and reduce precision loss.
- Because it has structured numerical features, moderate size, and low input dimensionality, it is efficient to train and deploy a compact DNN on devices with constrained resources.
- I would choose the QAT INT8 model because it has a small model size and high accuracy, which is good for devices with limited memory and compute.

Knowledge Distillation with Pruning and Quantization



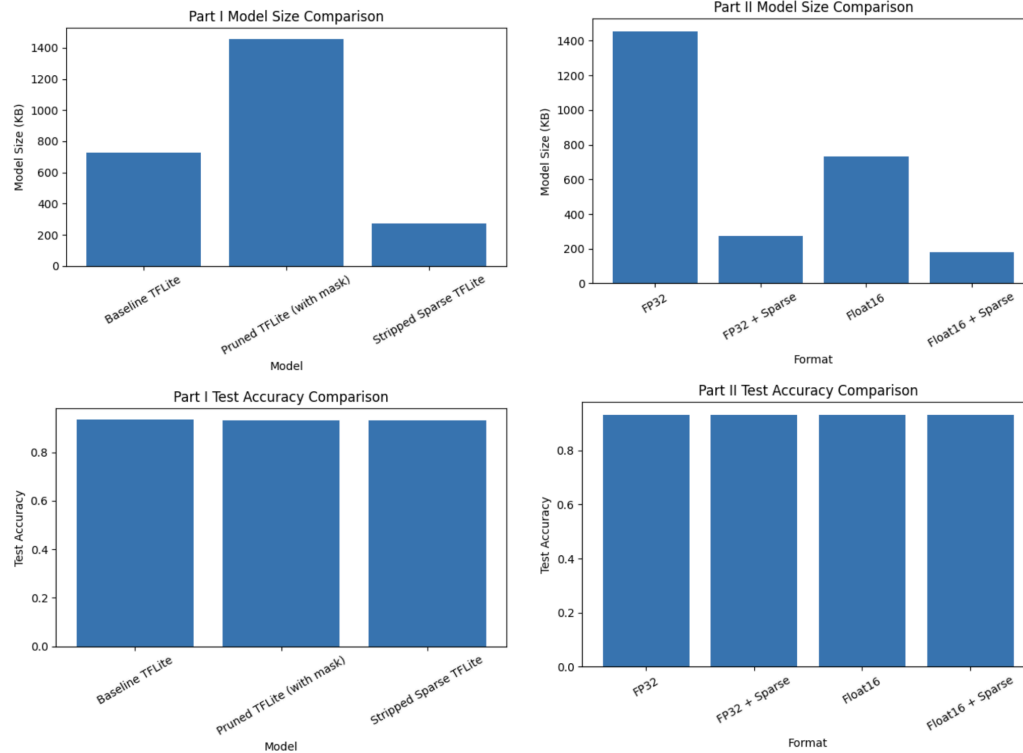
	Model	Format	Test Accuracy	Model Size (KB)
0	Distilled Student TFLite	FP32	0.939600	316.886719
1	Pruned Distilled TFLite (with mask)	FP32	0.946386	634.687500
2	Stripped Sparse Distilled TFLite	FP32 + Sparse	0.946386	127.984375
3	Stripped Sparse Distilled TFLite	INT8 + Sparse	0.946725	65.531250

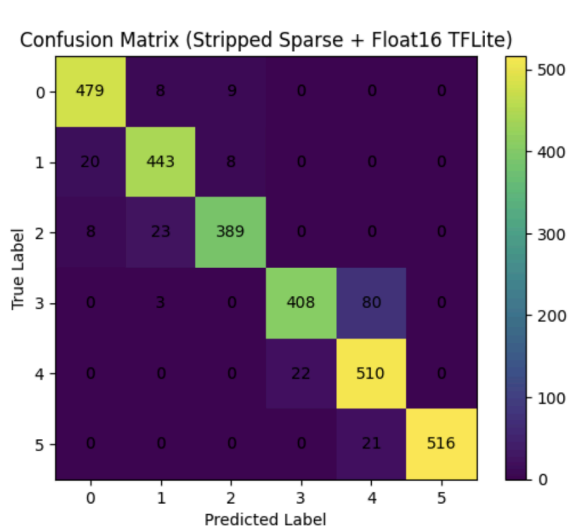
Short answer:

- The distilled student has higher accuracy despite having the same architecture. The distilled model was learned more effectively by leveraging information from the teacher model.
- Yes, because knowledge distillation helped the smaller model retain performance by transferring soft target information from the teacher, which allowed the student to capture the class relationships and improve generalization.
- Pruning reduced the number of active weights, but the model size only decreased significantly after stripping and applying sparse optimizations. INT8 quantization reduced the model size by converting weights and activations from 32-bit to 8-bit and reduced memory.

4. I chose a stripped sparse INT8 model because it has a very small model size and good accuracy.
5. The final sparse INT8 model is ideal because it combines quantization, sparsity, and distribution, which reduces memory and computation, while preserving accuracy. This model is compact, efficient, and accurate.

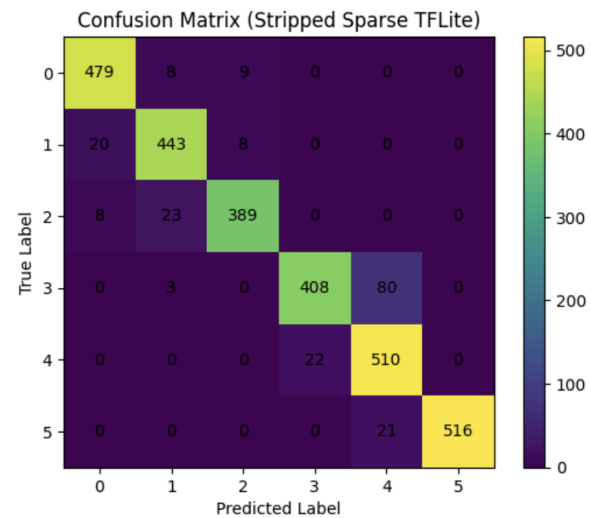
Pruning and Quantization of a DNN Using the UCI Human Activity Recognition Dataset





Classification Report (Stripped Sparse + Float16 TFLite):

	precision	recall	f1-score	support
0	0.94	0.97	0.96	496
1	0.93	0.94	0.93	471
2	0.96	0.93	0.94	420
3	0.95	0.83	0.89	491
4	0.83	0.96	0.89	532
5	1.00	0.96	0.98	537
accuracy			0.93	2947
macro avg	0.94	0.93	0.93	2947
weighted avg	0.93	0.93	0.93	2947



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weighted avg	0.93	0.93	0.93	2947

Summary:

1. No, pruning alone didn't reduce file size significantly because the masked weights are still stored in dense form.
2. It removes the pruning masks and wrapper layers, and leaves only the compressed sparse weights, which enables actual size reduction and optimization.
3. No, it didn't change the test accuracy even though it reduces model size.
4. I chose the stripped sparse INT16 model because it has minimal memory usage and high accuracy.